

目 次

|           |                   |       |     |
|-----------|-------------------|-------|-----|
|           | 伏図                | ..... | 1   |
| 新 築 工 事   | 荷重の算定             | ..... | 2   |
|           | 壁柱の軸力・地震力及び鉄筋量の算定 | ..... | 4   |
| 構 造 計 算 書 | たわみの算定            | ..... | 1 1 |
|           | 基礎の設計             | ..... | 1 2 |

平成 1 5 年 9 月

Translation

Title Page / Table of Contents

Project Type

New Construction

Document Title

Structural Calculation Report

Table of Contents

|                                                                                       |    |
|---------------------------------------------------------------------------------------|----|
| 1. Foundation Plan (Layout Drawing) .....                                             | 1  |
| 2. Load Calculation .....                                                             | 2  |
| 3. Wall and Column Axial Forces,<br>Seismic Forces, and Reinforcement Calculation ... | 4  |
| 4. Deflection Calculation .....                                                       | 11 |
| 5. Foundation Design .....                                                            | 12 |

Date

September, 2003

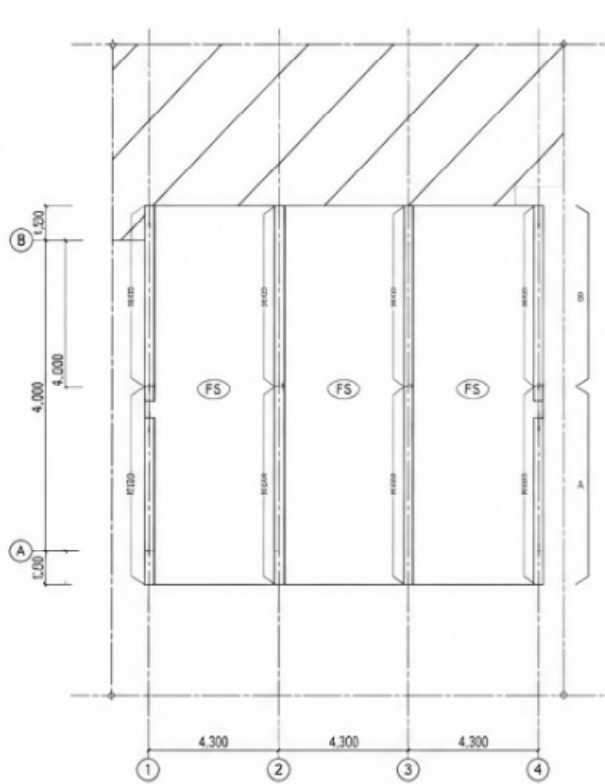
Readability Status

- Readable (confirmed)

- All text and numbers are clearly readable.
- Partially Difficult to Read

- None
- Not Readable / Unclear

- None



伏図 1:100

## &lt; 一般仕様 &gt;

## コンクリート

基礎・基礎梁  
普通コンクリート $F_c=21\text{N/mm}^2$ 

(比重 2.3)

## 上階コン

普通コンクリート

 $F_c=18\text{N/mm}^2$ 

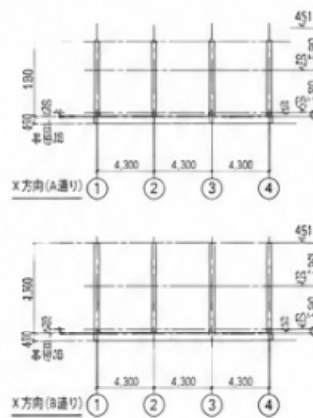
(比重 2.1)

鉄筋 SD295 普通鉄筋  
(JIS規格品)

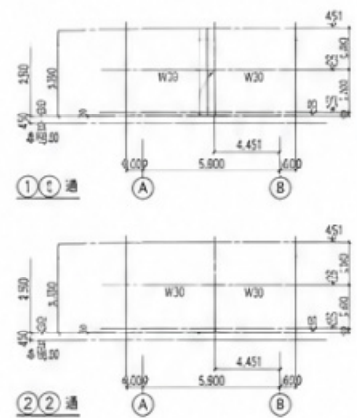
## 備考・本基礎は

布基礎設計図計 DA/0W  
布基礎以外は本設計でない場合は  
地盤改良を行う

## \*特記なき限り

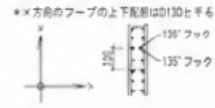
基礎底面高さ GL - 1.50  
基礎天端高さ GL - 0.45鉄筋コンクリート  
(D13 @200 (片筋))

軸図 1:100



| 記号   | 断面  | 鉄筋                           | 地中埋(フープ)  |
|------|-----|------------------------------|-----------|
| W30  | 300 | D16-φ150W                    | D10-φ100W |
| W30A | 300 | 上: D16-φ150W<br>下: D16-φ150W | D10-φ100W |

## 壁配筋要領

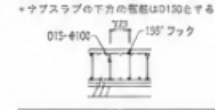


\* X方向のフープの上下配筋はD13とする

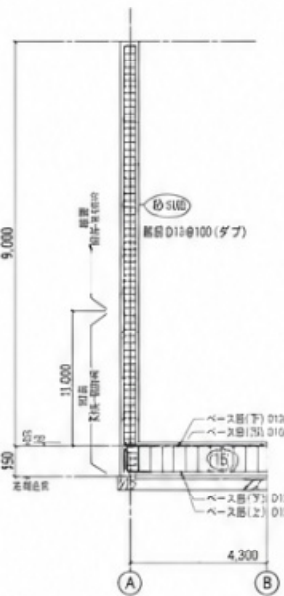


| 記号 | 断面  | X方向主筋(片筋) 上下       | Y方向主筋(片筋) スラブラリ    | 備考   |
|----|-----|--------------------|--------------------|------|
| FS | 600 | 上: D16W<br>下: D16W | 上: D13W<br>下: D13W | モザイク |

## 基礎配筋要領



\* スラブラリの下の配筋はD13とする



X方向(W30)部標準断面図 1/50

## Translation

## Drawings Included

- Plan View (Scale 1:100)
- Elevations / Frame Diagrams (Scale 1:100)
- Standard Section (W30) (Scale 1:50)

## General Specifications

- Concrete
  - Foundation, Foundation Beams: Ordinary Concrete  $F_c=21\text{ N/mm}^2$  (Unit weight 2.3)
  - Slab on Grade: Ordinary Concrete  $F_c=18\text{ N/mm}^2$  (Unit weight 2.1)
- Reinforcement: SD295, ordinary reinforcing steel (JIS standard)
- Notes:
  - This foundation follows the Strip Foundation Design Guideline DA/0V.
  - If other than strip foundation, ground improvement shall be performed when not covered by this guideline.

## Notes (unless otherwise specified)

- Foundation bottom level: GL - 1.50 m
- Foundation top level: GL - 0.45 m

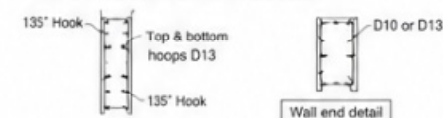
Hatched concrete: D13 @200 (one-way)

## Beam Schedule

| Symbol | Section | Main Bars                                 | Hoop      |
|--------|---------|-------------------------------------------|-----------|
| W30    | 300     | D16-φ150W                                 | D10-φ100W |
| W30A   | 300     | 上: D16-φ150W<br>下: D16-φ150W<br>D16-φ200W | D10-φ100W |

## Wall Reinforcement Details

- Hoops in X-direction: D13 at top and bottom



## Foundation Slab Reinforcement

| Symbol | Section | X-direction main bars (Perimeter) | Y-direction main bars (Interior) From slab above | Remarks |
|--------|---------|-----------------------------------|--------------------------------------------------|---------|
| FS     | 600     | Top 3-D16W<br>Bottom 3-D16W       | Top 3-D13W<br>Bottom 3-D13W                      | Mosaic  |

## Base Slab Reinforcement Details

- Bottom reinforcement below sub-slab: D13D



## Readability Status

## Readable (confirmed)

- All drawings, dimensions, symbols, and annotations are clearly readable.
- Tables and reinforcement details are legible.

## Partially Difficult to Read

- None

## Not Readable / Unclear

- None

< 荷重の算定 >

|      |           |      | D. L.        | 積雪    | 短期         | 長期換算<br>(梁設計用)                 |
|------|-----------|------|--------------|-------|------------|--------------------------------|
| ・屋根  | 銅板 (下地共)  | 0.20 | 0.6<br>(地震用) | 0.6   | 1.2        | /1.5→ 0.8<br>kN/m <sup>2</sup> |
|      | 母屋、小屋組    | 0.20 |              |       |            |                                |
|      | 天井        | 0.20 |              |       |            |                                |
| ・居室床 | 間仕切       | 0.15 | 0.8          | S 1.8 | 2.6        | kN/m <sup>2</sup>              |
|      | 木床組 (大引共) | 0.25 |              | R 1.3 | 2.1        |                                |
|      | 床梁        | 0.20 |              | E 0.6 | <u>1.4</u> |                                |
|      | 天井        | 0.20 |              |       |            |                                |

|             |          |      |            |                   |
|-------------|----------|------|------------|-------------------|
| ・外壁<br>(木造) | 銅板 (下地共) | 0.15 | <u>0.5</u> | kN/m <sup>2</sup> |
|             | 胴縁・間柱    | 0.20 |            |                   |
|             | 内装       | 0.15 |            |                   |

\*サイディング、サッシュ廻りもこれに準ずる

h = 1.2 → 0.5 kN/m  
h = 0.7 → 0.3 kN/m

|              |            |      |            |                   |
|--------------|------------|------|------------|-------------------|
| ・RC壁<br>(外壁) | コンクリート厚300 | 7.20 | <u>7.7</u> | kN/m <sup>2</sup> |
|              | 打増 (厚15)   | 0.35 |            |                   |
|              | 内装         | 0.15 |            |                   |
| ・RC壁<br>(内壁) | コンクリート厚300 | 7.20 | <u>7.5</u> | kN/m <sup>2</sup> |
|              | 内装         | 0.30 |            |                   |

\* 地震時 Co = 0.3 とする。

Translation

< Calculation of Loads >

| Calculation of Loads >                             |      |   | D.L.                     | Snow Load         | Short Term        | Long Term Conversion<br>(For beam design) |
|----------------------------------------------------|------|---|--------------------------|-------------------|-------------------|-------------------------------------------|
|                                                    |      |   |                          |                   |                   |                                           |
| • Roof                                             |      |   |                          |                   |                   |                                           |
| Steel plate (including substrate)                  | 0.20 | } | 0.6<br>(For seismic use) | 0.6               | 1.2               | /1.5→ 0.8<br>kN/m <sup>2</sup>            |
| Purlins, roof framing                              | 0.20 |   |                          |                   |                   |                                           |
| Ceiling                                            | 0.20 |   |                          |                   |                   |                                           |
|                                                    |      |   |                          |                   |                   |                                           |
| • Floor (Living Area)                              |      |   |                          |                   |                   |                                           |
| Partitions                                         | 0.15 | } | 0.8                      | S<br>R<br>E       | 1.8<br>1.3<br>0.6 | 2.6<br>2.1<br><u>1.4</u>                  |
| Wood floor framing (including joists)              | 0.25 |   |                          |                   |                   |                                           |
| Floor beams                                        | 0.20 |   |                          |                   |                   |                                           |
| Ceiling                                            | 0.20 |   |                          |                   |                   |                                           |
|                                                    |      |   |                          |                   |                   |                                           |
| • Exterior Wall<br>(Wooden Construction)           |      |   |                          |                   |                   |                                           |
| Steel plate (including substrate)                  | 0.15 | } | 0.5                      | kN/m <sup>2</sup> |                   |                                           |
| Furring strips / studs                             | 0.20 |   |                          |                   |                   |                                           |
| Interior finish                                    | 0.15 |   |                          |                   |                   |                                           |
| * Applies also to siding and around windows/doors. |      |   |                          |                   |                   |                                           |
| h = 1.2 → 0.5 kN/m                                 |      |   |                          |                   |                   |                                           |
| h = 0.7 → 0.3 kN/m                                 |      |   |                          |                   |                   |                                           |
|                                                    |      |   |                          |                   |                   |                                           |
| • RC Wall<br>(Concrete)                            |      |   |                          |                   |                   |                                           |
| Concrete thickness 300                             | 7.20 |   | <u>7.7</u>               | kN/m <sup>2</sup> |                   |                                           |
| Additional concrete (thickness 15)                 | 0.35 |   |                          |                   |                   |                                           |
| Interior finish                                    | 0.15 |   |                          |                   |                   |                                           |
|                                                    |      |   |                          |                   |                   |                                           |
| • RC Wall<br>(Concrete)                            |      |   |                          |                   |                   |                                           |
| Concrete thickness 300                             | 7.20 |   | <u>7.5</u>               | kN/m <sup>2</sup> |                   |                                           |
| Interior finish                                    | 0.30 |   |                          |                   |                   |                                           |

\* During earthquakes, Co = 0.3 is used.

Readability Status

☒ Readable (confirmed)

- All text and numbers are clearly readable.

☐ Partially Difficult to Read

- None

☐ Not Readable / Unclear

- None

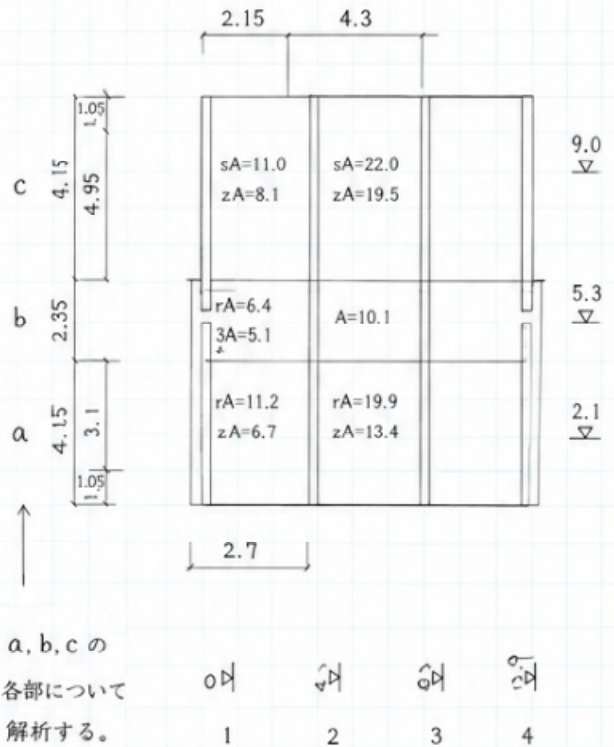


< 柱軸力算定用データ >

( ) 内は地震用

|   | 床 (kN/m <sup>2</sup> ) |              |   | 梁等 (kN/m) |  |  |  |
|---|------------------------|--------------|---|-----------|--|--|--|
|   | a                      | b            | c |           |  |  |  |
| R | 0.8<br>(0.6)           | 0.8<br>(0.6) |   |           |  |  |  |
| 3 | 2.1<br>(1.4)           | 0.8<br>(0.6) |   |           |  |  |  |
| 2 | 2.1<br>(1.4)           | 2.1<br>(1.4) |   |           |  |  |  |
| 1 |                        |              |   |           |  |  |  |

|   | 階高<br>(m)      | 壁 (kN/m <sup>2</sup> ) |       |       | 柱 (kN/m) |  |  |
|---|----------------|------------------------|-------|-------|----------|--|--|
|   |                | RC. 9                  | RC. 内 | 木(内外) |          |  |  |
| 3 | b:1.6          | 7.7                    | 7.5   | 0.5   |          |  |  |
| 2 | b:3.4<br>c:2.6 | "                      | "     | "     |          |  |  |
| 1 | 3.1            | "                      | "     | "     |          |  |  |



- \* 柱軸力算定プログラムについて
1. 上記の単位荷重及び、柱ごとの各部材数量を入力し、その積を集計することによって軸力を算定する。
  2. PH等の特殊荷重はその他付加荷重としてデータを入力する。
  3. また、柱座標を入力することにより階ごとの重心の座標も算定する。

\* 計算用剖心寸法  
及び 柱負担床面積表

Translation

< Data for Column Axial Force Calculation >  
( ) indicates seismic load

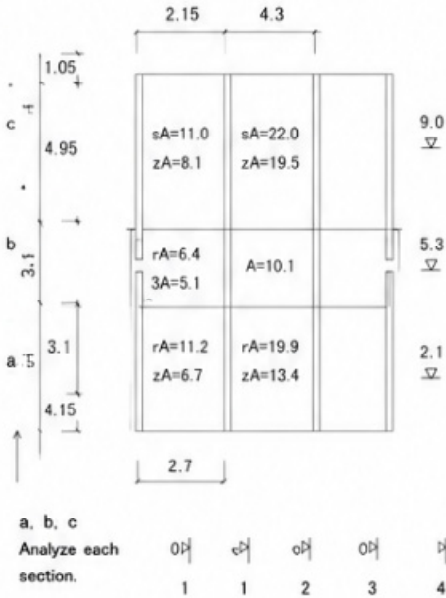
|   | Floor (kN/m <sup>2</sup> ) |              |   | Beam, etc. (kN/m) |  |  |  |
|---|----------------------------|--------------|---|-------------------|--|--|--|
|   | a                          | b            | c |                   |  |  |  |
| R | 0.8<br>(0.6)               | 0.8<br>(0.6) |   |                   |  |  |  |
| 3 | 2.1<br>(1.4)               | 0.8<br>(0.6) |   |                   |  |  |  |
| 2 | 2.1<br>(1.4)               | 2.1<br>(1.4) |   |                   |  |  |  |
| 1 |                            |              |   |                   |  |  |  |

|   | Story Height (m) | Wall (kN/m <sup>2</sup> ) |              |                | Column (kN/m) |  |  |
|---|------------------|---------------------------|--------------|----------------|---------------|--|--|
|   |                  | RC. 9                     | RC. Internal | Wood (Int/Ext) |               |  |  |
| 3 | b:1.6            | 7.7                       | 7.5          | 0.5            |               |  |  |
| 2 | b:3.4<br>c:2.6   | -                         | -            | -              |               |  |  |
| 1 | 3.1              | -                         | -            | -              |               |  |  |

- \* About the column axial force calculation program
1. The axial force is calculated by inputting the above unit loads and the quantity of each member for each column, and aggregating their products.
  2. Special loads such as PH, etc. are input as additional loads.
  3. Also, by inputting column coordinates, the centroid coordinates for each story are also calculated.

\* Calculation section dimensions  
and column-supported floor area table



Readability Status

☒ Readable (confirmed)

- All text and numbers are clearly readable.

☐ Partially Difficult to Read

- None

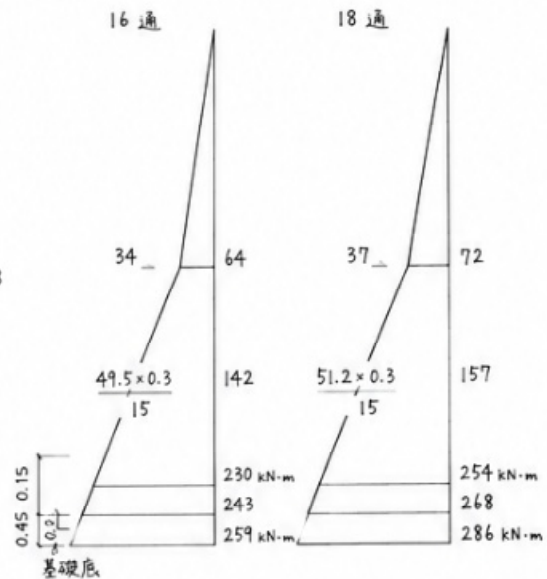
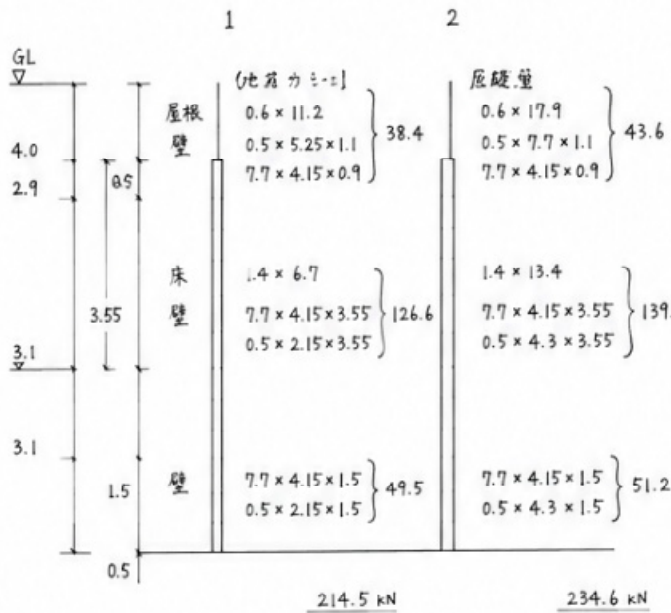
☐ Not Readable / Unclear

- None

## &lt; a 部の荷重 &gt;

(地震用質量)

(モーメント図) 水平力はPS 参照



$$\text{積載の補正} \quad (0.8 - 0.6) \times 11.2$$

$$(2.1 - 1.4) \times 6.7$$

$$(0.8 - 0.6) \times 17.9$$

$$(2.1 - 1.4) \times 13.4$$

壁 1m あたりのモーメント

$$230 / 4.15 = 55 \text{ kN-m/m}$$

$$254 / 4.15 = 61 \text{ kN-m/m}$$

$$\text{壁脚軸力} \quad 221.4 \text{ kN}$$

$$247.6 \text{ kN}$$

$$D = 300$$

$$d = 240$$

$$j = 210$$

$$l_f = 295$$

$$M / f_{tj} = 888 \text{ mm}^2$$

^

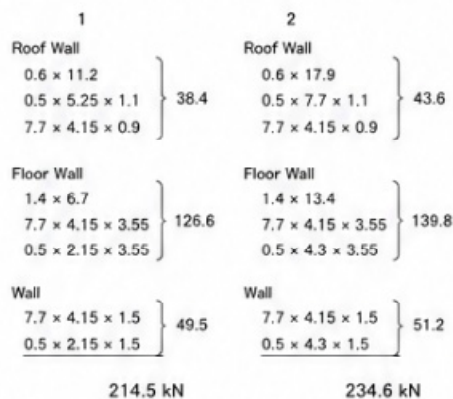
$$D16 - \phi 160$$

$$(a_t = 1243 \text{ mm}^2)$$

$$M / f_{tj} = 985 \text{ mm}^2$$

## Translation

## &lt; Load on Section a &gt; (Seismic Mass)



## Live Load Correction

$$(0.8 - 0.6) \times 11.2$$

$$(2.1 - 1.4) \times 6.7$$

$$(0.8 - 0.6) \times 17.9$$

$$(2.1 - 1.4) \times 13.4$$

## Axial Force at Wall Base

$$221.4 \text{ kN}$$

$$247.6 \text{ kN}$$

$$D = 300$$

$$d = 240$$

$$j = 210$$

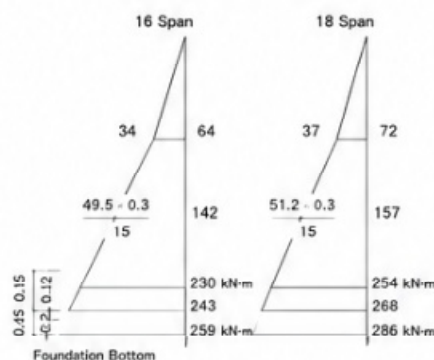
$$l_f = 295$$

$$M / f_{tj} = 888 \text{ mm}^2 \quad M / f_{tj} = 985 \text{ mm}^2$$

^

$$D16 - \phi 160$$

$$(a_t = 1243 \text{ mm}^2)$$

(Moment Diagram)  
Horizontal force refers to PS.

## Moment per 1 m of Wall

$$230 / 4.15 = 55 \text{ kN-m/m}$$

$$254 / 4.15 = 61 \text{ kN-m/m}$$

## Readability Status

○ Readable (confirmed)

- All main calculations, diagrams, and values are clearly readable.

○ Partially Difficult to Read

- Handwritten symbols are slightly unclear in some places.

○ Not Readable / Unclear

- None affecting structural interpretation.

< a-1 >

\* 地震力の算定 \*

H= 7.1    T =0.153    Tc=0.60    2T/(1+3T)=0.210  
Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 38  | 38  | 0.233 | 1.386 | 0.416 | 16 | 16 |
| 2 | 127 | 165 | 1.000 | 1.000 | 0.300 | 50 | 34 |

< a-2 >

\* 地震力の算定 \*

H= 7.1    T =0.153    Tc=0.60    2T/(1+3T)=0.210  
Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 44  | 44  | 0.238 | 1.380 | 0.414 | 18 | 18 |
| 2 | 140 | 183 | 1.000 | 1.000 | 0.300 | 55 | 37 |

< a-1 >

• Seismic Force Calculation

H = 7.1    T = 0.153    Tc = 0.60    2T/(1+3T) = 0.210  
Z = 1.0    Rt = 1.000    Co = 0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 38  | 38  | 0.233 | 1.386 | 0.416 | 16 | 16 |
| 2 | 127 | 165 | 1.000 | 1.000 | 0.300 | 50 | 34 |

< a-2 >

• Seismic Force Calculation

H = 7.1    T = 0.153    Tc = 0.60    2T/(1+3T) = 0.210  
Z = 1.0    Rt = 1.000    Co = 0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 44  | 44  | 0.238 | 1.380 | 0.414 | 18 | 18 |
| 2 | 140 | 183 | 1.000 | 1.000 | 0.300 | 55 | 37 |

 Readable (confirmed)

- All text, numbers, and table values are clearly readable.

 Partially Difficult to Read

- None

 Not Readable / Unclear

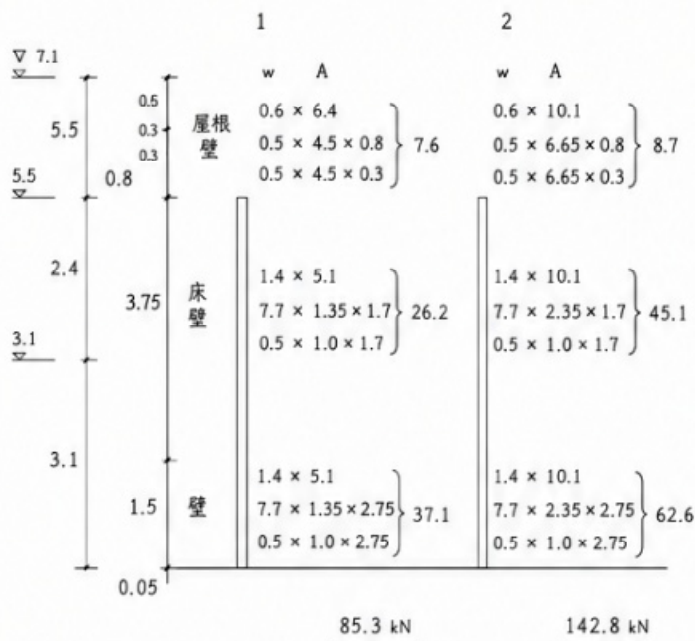
- None



## &lt; b 部の荷重 &gt;

(地震時慣性量)

(モーメント図) 水平力は P 7 参照.

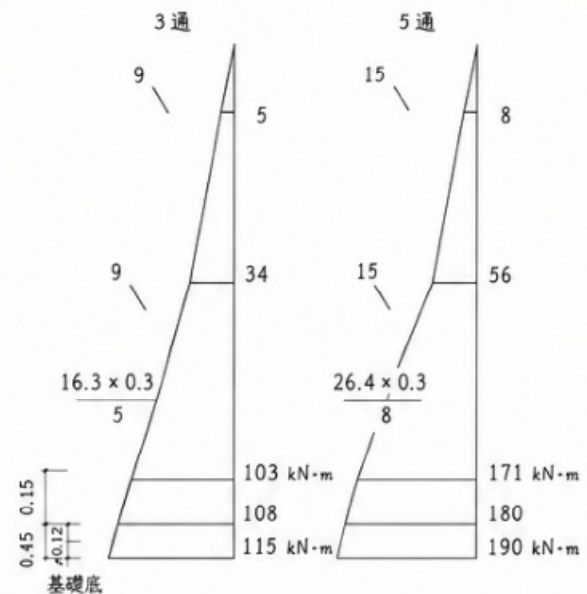


積載の補正

| Section | Correction                                             |
|---------|--------------------------------------------------------|
| 1       | $(0.8 - 0.6) \times 6.4$<br>$(2.1 - 1.4) \times 5.1$   |
| 2       | $(0.8 - 0.6) \times 10.1$<br>$(2.1 - 1.4) \times 10.1$ |

壁脚軸力

| Section | Value (kN) |
|---------|------------|
| 1       | 93.7       |
| 2       | 159.0      |



壁 1 m あたりのモーメント

Moment per 1 m of Wall

| Section | Value (kN-m/m)    |
|---------|-------------------|
| 1       | $103 / 1.35 = 76$ |
| 2       | $171 / 2.35 = 73$ |

Design Parameters

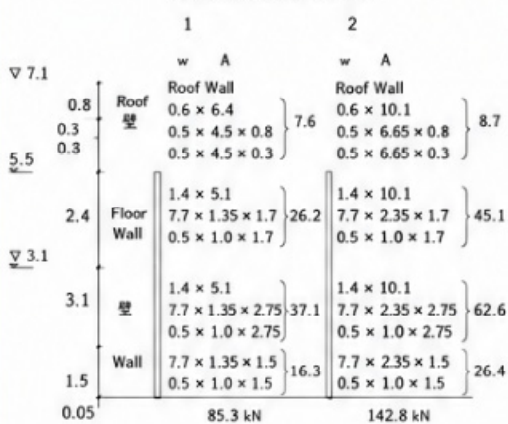
| Parameter     | Value                            |
|---------------|----------------------------------|
| D             | 300                              |
| d             | 240                              |
| j             | 210                              |
| $f_t$         | 295                              |
| $M / f_{tj}$  | 1226 mm <sup>2</sup>             |
| $M / f_{tj}$  | 1178 mm <sup>2</sup>             |
| Reinforcement | D16 - $\phi$ 160                 |
| Area          | ( $A_t = 1243$ mm <sup>2</sup> ) |

6

## Translation

## &lt; Load on Section b &gt;

(Seismic Inertial Weight)



Live Load Correction

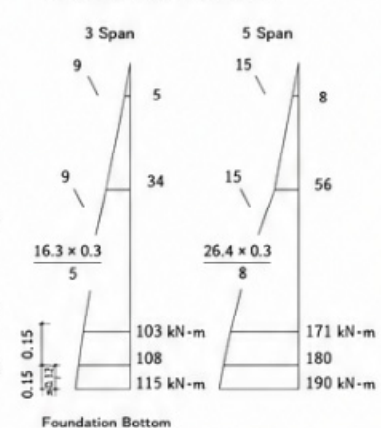
| Section | Correction                                             |
|---------|--------------------------------------------------------|
| 1       | $(0.8 - 0.6) \times 6.4$<br>$(2.1 - 1.4) \times 5.1$   |
| 2       | $(0.8 - 0.6) \times 10.1$<br>$(2.1 - 1.4) \times 10.1$ |

Axial Force at Wall Base

| Section | Value (kN) |
|---------|------------|
| 1       | 93.7       |
| 2       | 159.0      |

(Moment Diagram)

Horizontal force refers to P7.



Moment per 1 m of Wall

| Section | Value (kN-m/m)    |
|---------|-------------------|
| 1       | $103 / 1.35 = 76$ |
| 2       | $171 / 2.35 = 73$ |

Design Parameters

| Parameter     | Value                            |
|---------------|----------------------------------|
| D             | 300                              |
| d             | 240                              |
| j             | 210                              |
| $f_t$         | 295                              |
| $M / f_{tj}$  | 1226 mm <sup>2</sup>             |
| $M / f_{tj}$  | 1178 mm <sup>2</sup>             |
| Reinforcement | D16 - $\phi$ 160                 |
| Area          | ( $A_t = 1243$ mm <sup>2</sup> ) |

## Readability Status

☒ Readable (confirmed)

- All main calculations, diagrams, and values are clearly readable.

☐ Partially Difficult to Read

- Handwritten symbols are slightly unclear in some places.

☐ Not Readable / Unclear

- None affecting structural interpretation.

&lt; b-1 &gt;

## \* 地震力の算定 \*

H= 7.1      T =0.153      Tc=0.60       $2T/(1+3T)=0.210$   
 Z= 1.0      Rt=1.000      Co=0.30

| F | W  | ΣW | α     | Ai    | Ci    | Q  | P |
|---|----|----|-------|-------|-------|----|---|
| 4 | 6  | 6  | 0.081 | 1.719 | 0.516 | 3  | 3 |
| 3 | 26 | 32 | 0.462 | 1.211 | 0.363 | 12 | 9 |
| 2 | 37 | 69 | 1.000 | 1.000 | 0.300 | 21 | 9 |

&lt; b-2 &gt;

## \* 地震力の算定 \*

H= 7.1      T =0.153      Tc=0.60       $2T/(1+3T)=0.210$   
 Z= 1.0      Rt=1.000      Co=0.30

| F | W  | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|----|-----|-------|-------|-------|----|----|
| 4 | 9  | 9   | 0.075 | 1.751 | 0.525 | 5  | 5  |
| 3 | 45 | 54  | 0.462 | 1.212 | 0.363 | 20 | 15 |
| 2 | 63 | 116 | 1.000 | 1.000 | 0.300 | 35 | 15 |

## Translation

## • (b-1)

## \* Calculation of Seismic Force \*

H= 7.1      T =0.153      Tc=0.60       $2T/(1+3T)=0.210$   
 Z= 1.0      Rt=1.000      Co=0.30

| F | W  | ΣW | α     | Ai    | Ci    | Q  | P |
|---|----|----|-------|-------|-------|----|---|
| 4 | 6  | 6  | 0.081 | 1.719 | 0.516 | 3  | 3 |
| 3 | 26 | 32 | 0.462 | 1.211 | 0.363 | 12 | 9 |
| 2 | 37 | 69 | 1.000 | 1.000 | 0.300 | 21 | 9 |

## • (b-2)

## \* Calculation of Seismic Force \*

H= 7.1      T =0.153      Tc=0.60       $2T/(1+3T)=0.210$   
 Z= 1.0      Rt=1.000      Co=0.30

| F | W  | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|----|-----|-------|-------|-------|----|----|
| 4 | 9  | 9   | 0.075 | 1.751 | 0.525 | 5  | 5  |
| 3 | 45 | 54  | 0.462 | 1.212 | 0.363 | 20 | 15 |
| 2 | 63 | 116 | 1.000 | 1.000 | 0.300 | 35 | 15 |

## Readability Status

 Readable (confirmed)

- All text and numbers are clearly readable.

 Partially Difficult to Read

- None

 Not Readable / Unclear

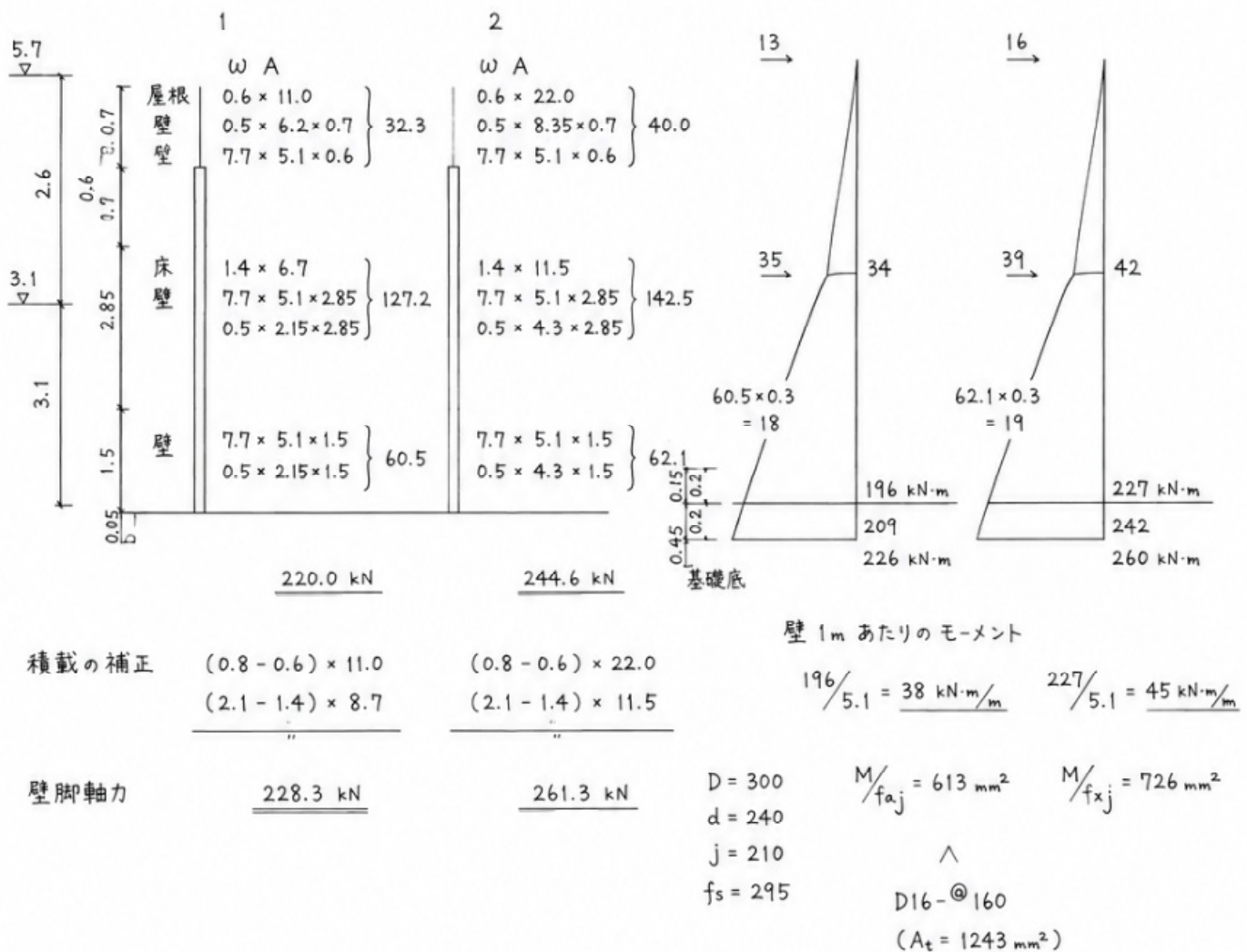
- None



## &lt; C 部の荷重 &gt;

(地震用層重量)

(モーメント図) 水平力は P9 参照.



## Translation

- Loads for Part "c"  
(Seismic story weight)

| 1                | 2                |
|------------------|------------------|
| Roof, wall       | Roof, wall       |
| 0.6 × 11.0       | 0.6 × 22.0       |
| 0.5 × 6.2 × 0.7  | 0.5 × 8.35 × 0.7 |
| 7.7 × 5.1 × 0.6  | 7.7 × 5.1 × 0.6  |
| 32.3             | 40.0             |
| Floor, wall      | Floor, wall      |
| 7.7 × 5.1 × 1.5  | 7.7 × 5.1 × 1.5  |
| 0.5 × 2.15 × 1.5 | 0.5 × 4.3 × 1.5  |
| 60.5             | 62.1             |
| 220.0 kN         | 244.6 kN         |

- Load correction

|                           |                           |
|---------------------------|---------------------------|
| $(0.8 - 0.6) \times 11.0$ | $(0.8 - 0.6) \times 22.0$ |
| $(2.1 - 1.4) \times 8.7$  | $(2.1 - 1.4) \times 11.5$ |
| " "                       | " "                       |

- Wall base axial force

|          |          |
|----------|----------|
| 228.3 kN | 261.3 kN |
|----------|----------|

- (Moment Diagram)  
Refer to P.9 for horizontal force.

| 13                            | 16                            |
|-------------------------------|-------------------------------|
| At top: 35                    | At top: 39                    |
| Mid: 34                       | Mid: 42                       |
| Lower: $60.5 \times 0.3 = 18$ | Lower: $62.1 \times 0.3 = 19$ |
| Base: 196 kN·m                | Base: 227 kN·m                |
| 209                           | 242                           |
| 226 kN·m                      | 260 kN·m                      |

- Moment per 1 m of wall  
 $196 / 5.1 = 38 \text{ kN·m/m}$   
 $227 / 5.1 = 45 \text{ kN·m/m}$

|          |                                          |
|----------|------------------------------------------|
| D = 300  | M/f <sub>aj</sub> = 613 mm <sup>2</sup>  |
| d = 240  | M/f <sub>xj</sub> = 726 mm <sup>2</sup>  |
| j = 210  | Λ                                        |
| fs = 295 | D16-@160                                 |
|          | (A <sub>t</sub> = 1243 mm <sup>2</sup> ) |

## Readability Status

☒ Readable (confirmed)

- All text and numbers are clearly readable.

☐ Partially Difficult to Read

- None

☐ Not Readable / Unclear

- None

&lt; c-1 &gt;

\* 地震力の算定 \*

H= 5.7    T =0.121    Tc=0.60     $2T/(1+3T)=0.178$   
 Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 32  | 32  | 0.203 | 1.359 | 0.408 | 13 | 13 |
| 2 | 127 | 160 | 1.000 | 1.000 | 0.300 | 48 | 35 |

&lt; c-2 &gt;

\* 地震力の算定 \*

H= 5.7    T =0.121    Tc=0.60     $2T/(1+3T)=0.178$   
 Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 40  | 40  | 0.219 | 1.340 | 0.402 | 16 | 16 |
| 2 | 143 | 183 | 1.000 | 1.000 | 0.300 | 55 | 39 |

## Translation

## • (c-1)

\* Calculation of Seismic Force \*

H= 5.7    T =0.121    Tc=0.60     $2T/(1+3T)=0.178$   
 Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 32  | 32  | 0.203 | 1.359 | 0.408 | 13 | 13 |
| 2 | 127 | 160 | 1.000 | 1.000 | 0.300 | 48 | 35 |

## • (c-2)

\* Calculation of Seismic Force \*

H= 5.7    T =0.121    Tc=0.60     $2T/(1+3T)=0.178$   
 Z= 1.0    Rt=1.000    Co=0.30

| F | W   | ΣW  | α     | Ai    | Ci    | Q  | P  |
|---|-----|-----|-------|-------|-------|----|----|
| 3 | 40  | 40  | 0.219 | 1.340 | 0.402 | 16 | 16 |
| 2 | 143 | 183 | 1.000 | 1.000 | 0.300 | 55 | 39 |

## Readability Status

 Readable (confirmed)

- All text and numbers are clearly readable.

 Partially Difficult to Read

- None

 Not Readable / Unclear

- None

## &lt; 壁柱について &gt;

柱の仕様を満足させるものとする.

軸筋比 0.8% として D16 を使用すると

$$300 \times 0.008 \times a < 2 \times 199 \text{ より (W30)}$$

$$a < 165.8 \rightarrow \underline{\text{D16}-\phi 160 \text{ W とする.}}$$

フープ  $\phi$  D10- $\phi$ 100 とすれば

1 本おき  $\phi$  320 にサブフープを設ければよい

$$P_{ws} = \frac{11.7}{320 \times 100} = 0.0022 > \underline{0.3\%}$$

但し a・b 部は

安全を考慮し GL+2.00 までは D16- $\phi$ 80 W とする.

(W30A)

$$D = 300$$

$$d = 240 \quad \delta = 210 \quad f_s = 295$$

$$A_t = 1243 \text{ mm}^2/\text{m}$$

$$M_g = A_t \cdot f_{tj} = 77.0 \text{ kN}\cdot\text{m.}$$

## Translation

- About wall columns

The column shall satisfy the required specifications.

Assuming a longitudinal reinforcement ratio of 0.8% and using D16,

$$300 \times 0.008 \times a < 2 \times 199 \quad (\text{W30})$$

$$a < 165.8 \rightarrow \underline{\text{Use D16-}\phi 160 \text{ W.}}$$

If hoops of  $\phi$  D10- $\phi$ 100 are used, provide a sub-hoop at every other  $\phi$ 320.

$$P_{ws} = \frac{11.7}{320 \times 100} = 0.0022 > \underline{0.3\%}$$

However, for parts a and b,

for safety, use D16- $\phi$ 80 W up to GL + 2.00.

(W30A)

## Readability Status

 Readable (confirmed)

- All text and numbers are clearly readable.

 Partially Difficult to Read

- None

 Not Readable / Unclear

- None



## &lt;たわみの算定&gt;

$$\text{b 部} \quad I = \frac{1350 \times 300^3}{12} = 3037.5 \times 10^6 \text{ mm}^3$$

$$\text{ヤング係数} \quad E = 3.35 \times 10^4 \times \left(\frac{23}{24}\right)^2 \times \left(\frac{21}{60}\right)^{\frac{1}{3}} = 21700 \text{ N/mm}^2$$

$$\text{たわみ} \quad y = \frac{P \times \ell^3}{6EI} \times \left\{ -\left(\frac{b}{\ell}\right)^3 + 3 \times \left(\frac{b}{\ell}\right)^2 \right\} \quad \begin{array}{l} \ell : \text{材長} \\ b : \text{支持端からのキヨリ} \end{array}$$

P=

$$\frac{7.1}{\nabla} \rightarrow 3000 \text{ N}$$

$$y = 3000 \times 0.00091 \times \left[ -\left(\frac{7.1}{11.1}\right)^3 + 3 \times \left(\frac{7.1}{11.1}\right)^2 \right] = 5.4$$

$$\Sigma y = 20.4 \text{ mm}$$

頂部最大変形

$$\frac{5.5}{\nabla} \rightarrow 9000 \text{ N}$$

$$y = 10.9$$

$$\Sigma y = 15.0 \text{ mm}$$

$$\frac{3.1}{\nabla} \rightarrow 9000 \text{ N}$$

$$y = 4.0$$

$$\Sigma y = 4.1 \text{ mm}$$

$$\frac{0.75}{\nabla} \rightarrow 5000 \text{ N}$$

$$y = 0.1 \text{ mm}$$



## Translation

## • Deflection Calculation

Part "b"

$$I = \frac{1350 \times 300^3}{12} = 3037.5 \times 10^6 \text{ mm}^3$$

Young's modulus:

$$E = 3.35 \times 10^4 \times \left(\frac{23}{24}\right)^2 \times \left(\frac{21}{60}\right)^{\frac{1}{3}} = 21700 \text{ N/mm}^2$$

Deflection:

$$y = \frac{P \times \ell^3}{6EI} \times \left\{ -\left(\frac{b}{\ell}\right)^3 + 3 \times \left(\frac{b}{\ell}\right)^2 \right\}$$

 $\ell$ : Member length

b: Distance from support end

P=

$$\frac{7.1}{\nabla} \rightarrow 3000 \text{ N}$$

$$y = 3000 \times 0.00091 \times \left[ -\left(\frac{7.1}{11.1}\right)^3 + 3 \times \left(\frac{7.1}{11.1}\right)^2 \right] = 5.4$$

$$\Sigma y = 20.4 \text{ mm}$$

Maximum deflection at top

$$\frac{5.5}{\nabla} \rightarrow 9000 \text{ N}$$

$$y = 10.9$$

$$\Sigma y = 15.0 \text{ mm}$$

$$\frac{3.1}{\nabla} \rightarrow 9000 \text{ N}$$

$$y = 4.0$$

$$\Sigma y = 4.1 \text{ mm}$$

$$\frac{0.75}{\nabla} \rightarrow 5000 \text{ N}$$

$$y = 0.1 \text{ mm}$$



## Readability Status



Readable (confirmed)

- All text and numbers are clearly readable.



Partially Difficult to Read

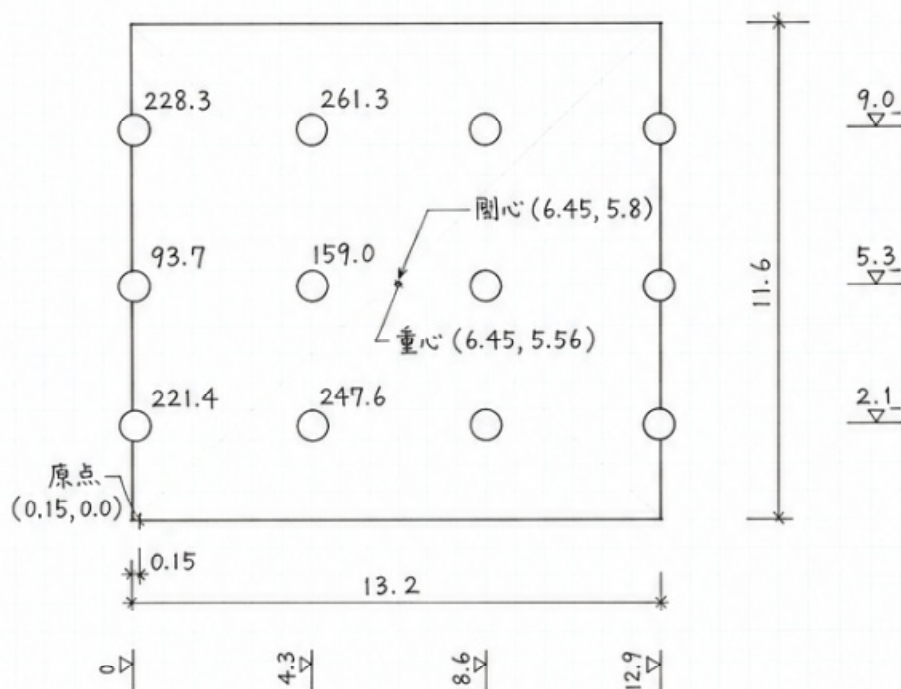
- None



Not Readable / Unclear

- None

## &lt; 基礎の設計 &gt;



12

## Translation

## &lt; Foundation Design &gt;

Foundation layout plan.

Column axial loads obtained from previous calculations are plotted on the foundation plan.

• Applied Column Axial Loads (kN)

| Position      | Axial Load |
|---------------|------------|
| Upper Left    | 228.3 kN   |
| Upper Center  | 261.3 kN   |
| Middle Left   | 93.7 kN    |
| Middle Center | 159.0 kN   |
| Lower Left    | 221.4 kN   |
| Lower Center  | 247.6 kN   |

• Center of Rigidity and Center of Gravity

Center of Rigidity : (6.45, 5.8)

Center of Gravity : (6.45, 5.56)

• Foundation Dimensions

Overall width : 13.2 m

Overall height : 11.6 m

• Reference Levels

GL +9.0 m

GL +5.3 m

GL +2.1 m

## Readability Status

○ Readable (confirmed)

- Foundation layout
- Column axial load values
- Center of rigidity coordinates
- Center of gravity coordinates
- Main dimensions
- Reference levels

○ Partially Difficult to Read

- Some handwritten annotations near the center coordinates
- Minor dimension notes

○ Not Readable / Unclear

- None affecting the structural interpretation

## &lt; ベタ基礎 地反力の算定 &gt;

|                          |                 |      |  |      |      |
|--------------------------|-----------------|------|--|------|------|
| C                        | 粘性土             |      |  |      |      |
| N = 3                    | 長期              |      |  | ic = | 1.00 |
| $\phi = 0.0$             | $\tan \theta =$ | 0.00 |  | iq = | 1.00 |
| C = 19 kN/m <sup>2</sup> | $\theta =$      | 0.00 |  | ir = | 0.00 |
| Df = 0.0 m               |                 |      |  |      |      |

|              |    |                   |         |                      |  |
|--------------|----|-------------------|---------|----------------------|--|
| $\gamma_2 =$ | 17 | kN/m <sup>3</sup> |         |                      |  |
| $\gamma_1 =$ | 7  | kN/m <sup>3</sup> |         |                      |  |
|              |    |                   | 長期設計地耐力 |                      |  |
| $\alpha a =$ | 37 | kN/m <sup>2</sup> | →       | 35 kN/m <sup>2</sup> |  |

|                       |      |               |                |      |                   |
|-----------------------|------|---------------|----------------|------|-------------------|
| 基礎底盤面積                | Af = | 153.1         | m <sup>2</sup> |      |                   |
|                       |      | (P4, 6, 8 より) |                |      |                   |
| 地上部建物重量               | Wo = | 2423          | kN             |      |                   |
| 一般基礎設計用反力 $\sigma_o'$ | =    | Wo / Af       | =              | 15.8 | kN/m <sup>2</sup> |

|             |      |      |                   |      |  |
|-------------|------|------|-------------------|------|--|
| 基礎単位面積あたり重量 |      |      |                   |      |  |
| 積載          |      | 1.8  |                   |      |  |
| 仕上等         |      | 0.4  |                   |      |  |
| 上間コン        | 24 x | =    | 0.0               |      |  |
| 土・基礎梁       | x    | =    | 0.0               |      |  |
| 底盤          | 24 x | 0.50 | =                 | 12.0 |  |
|             | wf = | 14.2 | kN/m <sup>2</sup> |      |  |

|                                              |              |                 |   |          |                   |
|----------------------------------------------|--------------|-----------------|---|----------|-------------------|
| 基礎重量                                         | Wf =         | wf × Af         | = | 2174     | kN                |
| 建物総重量 (基礎共)                                  | $\Sigma W =$ | Wo + Wf         | = | 4597     | kN                |
| 長期平均地反力度                                     | $\sigma_o =$ | $\Sigma W / Af$ | = | 30.0     | kN/m <sup>2</sup> |
| 長期最大地反力度                                     | Lo,max =     |                 | = | 32.0     | kN/m <sup>2</sup> |
| ( $\alpha \times \sigma_o \times \alpha a$ ) |              |                 |   | $\wedge$ |                   |
| 長期設計地耐力                                      | Lqa =        |                 | = | 35.0     | kN/m <sup>2</sup> |
|                                              |              |                 |   | O.K.     |                   |

|          |                   |          |                   |      |  |
|----------|-------------------|----------|-------------------|------|--|
| 短期最大地反力度 | $s\sigma_{max} =$ | 41.0     | kN/m <sup>2</sup> |      |  |
|          |                   | $\wedge$ |                   |      |  |
| 短期設計地耐力  | sqa =             | 70.0     | kN/m <sup>2</sup> |      |  |
|          |                   |          |                   | O.K. |  |

|             |       |       |              |  |  |
|-------------|-------|-------|--------------|--|--|
|             | X方向   | Y方向   |              |  |  |
| 基礎巾 L (m)   | 13.20 | 11.60 |              |  |  |
| 原点位置 O (m)  | 0.15  | 0.00  | (基礎左下隅からの距離) |  |  |
| 基礎圆心 J (m)  | 6.45  | 5.80  |              |  |  |
| 建物重心 g (m)  | 6.45  | 5.56  | ← (P 12 より)  |  |  |
| 偏心距離 eo (m) | 0.00  | 0.24  | ← J - g      |  |  |

## 長期

|                |   |     |           |  |  |
|----------------|---|-----|-----------|--|--|
| 偏心モーメント (kN・m) | 0 | 581 | ← eo × Wo |  |  |
| 土圧による回転モーメント   |   |     |           |  |  |
| 合計 LMe (kN・m)  | 0 | 581 |           |  |  |

|                             |       |       |                            |  |  |
|-----------------------------|-------|-------|----------------------------|--|--|
| 偏心距離 e (m)                  | 0.000 | 0.126 | ← LMe / $\Sigma W$         |  |  |
| e / L                       | 0.000 | 0.011 |                            |  |  |
| $\alpha$                    | 1.00  | 1.07  |                            |  |  |
| Lo,max (kN/m <sup>2</sup> ) | 30.0  | 32.0  | ← $\alpha \times \sigma_o$ |  |  |
| Lo,max (kN/m <sup>2</sup> ) | 15.8  | 17.8  | ← Lo,max - wf              |  |  |
|                             |       |       | (縁端部基礎設計用反力)               |  |  |

## 短期

|                                        |       |       |                            |  |  |
|----------------------------------------|-------|-------|----------------------------|--|--|
| 転倒モーメント sMo (kN・m)                     | 2672  | 2672  | ← (P4, 6, 8 より)            |  |  |
| 偏心モーメント sMe (kN・m)                     | 2672  | 3253  | ← LMe + sMo                |  |  |
| 偏心距離 e (m)                             | 0.58  | 0.71  | ← sMe / $\Sigma W$         |  |  |
| e / L                                  | 0.044 | 0.061 |                            |  |  |
| $\alpha$                               | 1.26  | 1.37  |                            |  |  |
| s $\sigma_{max}$ (kN/m <sup>2</sup> )  | 38.0  | 41.0  | ← $\alpha \times \sigma_o$ |  |  |
| s $\sigma_{omax}$ (kN/m <sup>2</sup> ) | 23.8  | 26.8  | ← s $\sigma_{max}$ - wf    |  |  |
| 短期 / 長期                                | 1.50  | 1.51  | (縁端部基礎設計用反力)               |  |  |

## Translation

## • Mat Foundation - Calculation of Soil Bearing Pressure

Soil Type: Cohesive Soil

|                            |                 |      |      |            |          |
|----------------------------|-----------------|------|------|------------|----------|
| • N = 3                    | Long Term       | ic = | 1.00 | Nc =       | 5.1      |
| • $\phi = 0.0$             | $\tan \theta =$ | 0.00 | iq = | 1.00       | Nq = 1.0 |
| • C = 19 kN/m <sup>2</sup> | $\theta =$      | 0.00 | ir = | 0.00       | Nr = 0.0 |
| • Df = 0.0 m               |                 |      |      | $\alpha =$ | 1.18     |

## • Foundation Area

Af = 153.1 m<sup>2</sup>

## • Superstructure Weight

Wo = 2423 kN (P4, 6, 8)

Average Design Pressure

 $\sigma_o' = Wo / Af = 15.8$  kN/m<sup>2</sup>

## • Weight per Unit Area

|                            |                        |
|----------------------------|------------------------|
| Live Load                  | 1.8                    |
| Finishes, etc.             | 0.4                    |
| Upper Floor Slab (24 cm)   | = 0.0                  |
| Soil & Foundation Beams    | = 0.0                  |
| Base Slab (24 cm × 0.50 m) | = 12.0                 |
| wf =                       | 14.2 kN/m <sup>2</sup> |

|                                     |              |                 |   |      |                   |
|-------------------------------------|--------------|-----------------|---|------|-------------------|
| • Foundation Weight                 | Wf =         | wf × Af         | = | 2174 | kN                |
| • Total Weight (with foundation)    | $\Sigma W =$ | Wo + Wf         | = | 4597 | kN                |
| • Average Soil Pressure (Long Term) | $\sigma_o =$ | $\Sigma W / Af$ | = | 30.0 | kN/m <sup>2</sup> |

## • Maximum Soil Pressure (Long Term)

(α × σ<sub>o</sub> × α<sub>a</sub>)

|                                        |       |      |                   |  |  |
|----------------------------------------|-------|------|-------------------|--|--|
| • Long-term Allowable Bearing Capacity | Lqa = | 35.0 | kN/m <sup>2</sup> |  |  |
|                                        |       | O.K. |                   |  |  |

## • Maximum Soil Pressure (Short Term)

s $\sigma_{max}$  = 41.0 kN/m<sup>2</sup>

|                                         |       |      |                   |  |  |
|-----------------------------------------|-------|------|-------------------|--|--|
| • Short-term Allowable Bearing Capacity | sqa = | 70.0 | kN/m <sup>2</sup> |  |  |
|                                         |       | O.K. |                   |  |  |

## X Direction Y Direction

|                             |       |       |  |  |                          |
|-----------------------------|-------|-------|--|--|--------------------------|
|                             | L (m) | 11.60 |  |  |                          |
| • Origin (O) (m)            | 0.15  | 0.00  |  |  | (from lower-left corner) |
| • Foundation Center (J) (m) | 6.45  | 5.80  |  |  |                          |
| • Building Center (g) (m)   | 6.45  | 5.56  |  |  | (From P12)               |
| • Eccentricity (eo) (m)     | 0.00  | 0.24  |  |  | (J - g)                  |

## Long Term

|                                        |       |       |                                    |  |  |
|----------------------------------------|-------|-------|------------------------------------|--|--|
| Eccentric Moment (kN・m)                | 0     | 581   | (eo × Wo)                          |  |  |
| Moment by Soil Pressure                |       |       |                                    |  |  |
| Total LMe (kN・m)                       | 0     | 581   |                                    |  |  |
| Eccentricity e (m)                     | 0.000 | 0.126 | (LMe / $\Sigma W$ )                |  |  |
| e / L                                  | 0.000 | 0.011 |                                    |  |  |
| $\alpha$                               | 1.00  | 1.07  |                                    |  |  |
| Lo,max (kN/m <sup>2</sup> )            | 30.0  | 32.0  | (α × σ <sub>o</sub> )              |  |  |
| s $\sigma_{omax}$ (kN/m <sup>2</sup> ) | 15.8  | 17.8  | (s $\sigma_{max}$ - wf)            |  |  |
|                                        |       |       | (Edge Bearing Pressure for Design) |  |  |

## Short Term

|                                        |       |       |                                    |  |  |
|----------------------------------------|-------|-------|------------------------------------|--|--|
| Overturning Moment sMo (kN・m)          | 2672  | 2672  | (P4, 6, 8)                         |  |  |
| Eccentric Moment sMe (kN・m)            | 2672  | 3253  | (LMe + sMo)                        |  |  |
| Eccentricity e (m)                     | 0.58  | 0.71  | (sMe / $\Sigma W$ )                |  |  |
| e / L                                  | 0.044 | 0.061 |                                    |  |  |
| $\alpha$                               | 1.26  | 1.37  |                                    |  |  |
| s $\sigma_{max}$ (kN/m <sup>2</sup> )  | 38.0  | 41.0  | (α × σ <sub>o</sub> )              |  |  |
| s $\sigma_{omax}$ (kN/m <sup>2</sup> ) | 23.8  | 26.8  | (s $\sigma_{max}$ - wf)            |  |  |
|                                        |       |       | (Edge Bearing Pressure for Design) |  |  |
| Short Term / Long Term                 | 1.50  | 1.51  |                                    |  |  |

## Readability Status

## ○ Readable (confirmed)

- All main calculation items and values are clearly readable.
- Dimensions, coordinates, and numerical results are identifiable.

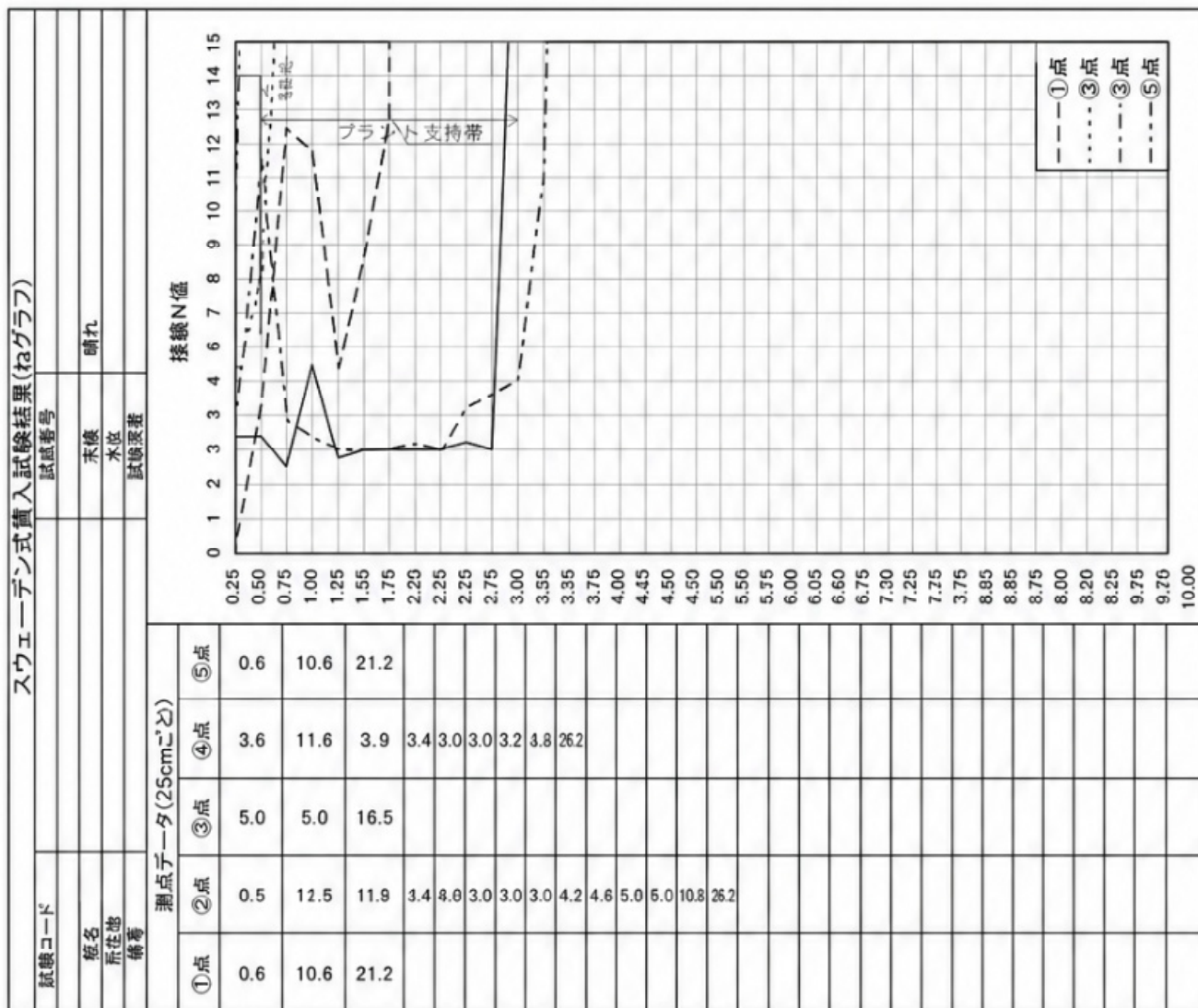
## ○ Partially Difficult to Read

- Handwritten note on the right side is slightly unclear.
- Some annotation references (e.g., "P12 より") may need context.

## ○ Not Readable / Unclear

- None affecting structural interpretation.





## Translation

- Swedish Weight Sounding Test Results (N-value Graph)

- Weather: Sunny
- Groundwater Level
- Test Depth
- Test Code
- Location Name
- Site Name
- Remarks

- Graph

- Vertical axis: N-value (blows / 0.25 m)
- Horizontal axis: Depth (m)

- Legend:

① Point      ② Point      ③ Point      ④ Point      ⑤ Point

- Note: Plant support zone

- Measurement Data (every 25 cm)

| Point   | N-value (Blows / 25 cm)                                            |
|---------|--------------------------------------------------------------------|
| ① Point | 3.4, 3.4, 2.5, 5.5, 2.8, 3.0, 3.0, 3.0, 3.0, 3.2, 23.0, 53.0       |
| ② Point | 0.5, 2.4, 12.5, 11.8, 4.4, 8.6, 12.8, 33.0                         |
| ③ Point | 5.0, 5.0, 16.5                                                     |
| ④ Point | 3.6, 11.6, 3.9, 3.4, 3.0, 3.0, 3.2, 3.0, 4.2, 4.6, 5.0, 10.8, 26.2 |
| ⑤ Point | 0.6, 10.6, 21.2                                                    |

### Readability Status

☐ Readable (confirmed)

- All text, numbers, and graph elements are clearly readable.

○ Partially Difficult to Read

- None

☐ Not Readable / Unclear

- None